

Indian Institute of
Technology Kharagpur

CS60111 GIS · TERM PROJECT · GROUP 8

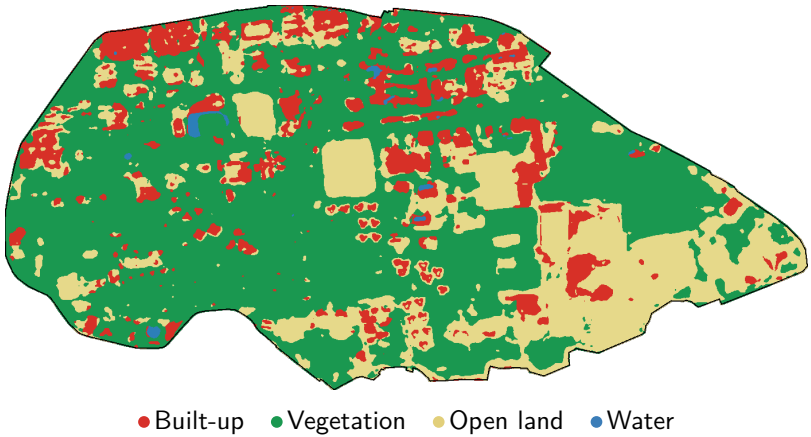
Land Use / Land Cover Mapping and Change Detection

IIT Kharagpur campus, 2020 vs 2025, from multi-temporal Sentinel-2

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Term project evaluation, Nov 2026

2025 land cover map, Random Forest, 10 m



Can we measure how our own campus changed?

RESEARCH QUESTION

How much of the 469 ha campus changed land cover between the 2019-20 and 2024-25 dry seasons, and from what to what?

Why a campus? Global products (Dynamic World, ESA WorldCover) map at 10 m but are tuned for regions. A 469 ha site tests whether they hold at the scale where planners actually decide: one construction compound, one lawn, one pond.

01 **Map 4 classes at 2 dates**
Sentinel-2, one Random Forest

02 **From-to change table in ha**
30 m minimum change width

03 **Accuracy against independent points**
WorldView-2 and -3, area-weighted

04 **Reproducible pipeline**
one Earth Engine script, one Python runner

New in this version: our own WorldView training labels, accuracy 0.67 → 0.74 (2020) and 0.77 → 0.80 (2025) · labels vs more data test · yearly dating of the south-east compound · is the change real? error-adjusted significance test · change flow diagram · gross gain, loss and swap per class · map area vs point-based area with 95% CI · model tuning log with a held-out half · feature importance · WorldView validation design

A 469 ha fenced campus, mapped at 10 m



Campus boundary: OpenStreetMap way 52435606 (184 vertices). Imagery: Google Earth, Airbus.

469.2 ha
MAIN FENCED CAMPUS

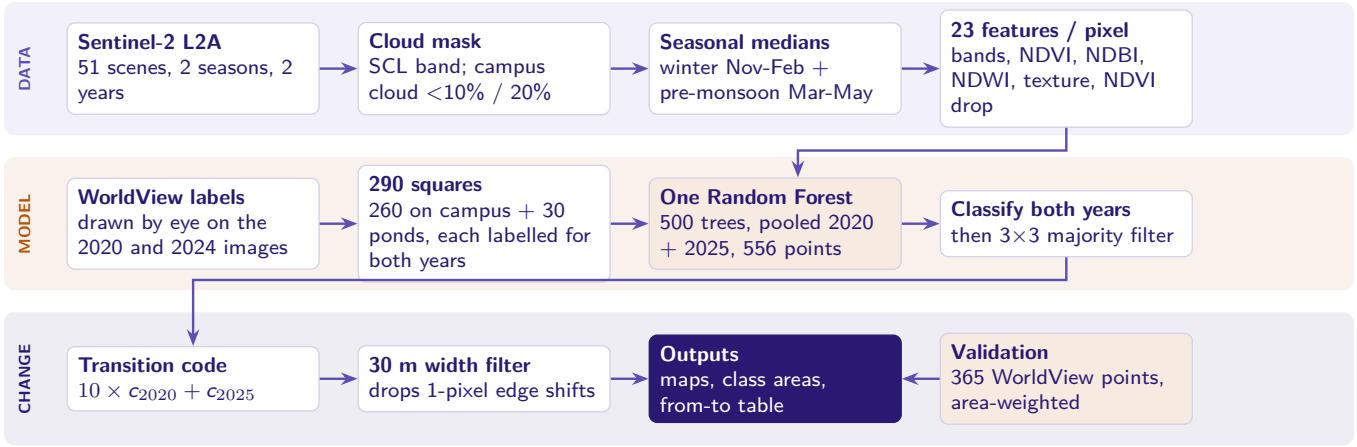
10 m
SENTINEL-2 PIXEL SIZE

45QWE
ONE TILE COVERS ALL OF IT

UTM 45N
EPSG:32645

Halls and quarters outside the fence are not in the study area.

From 51 satellite scenes to a from-to change table



One forest for both years, so an unchanged pixel gets the same rules in 2020 and 2025. Separate forests disagreed on 91 ha of unchanged roofs.

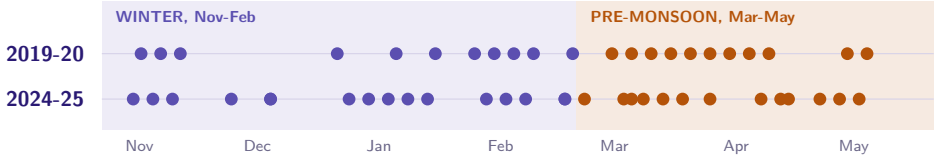
Same seasons in both years, so seasonal greening is not counted as change



Winter 2019-20 median composite, 11 scenes

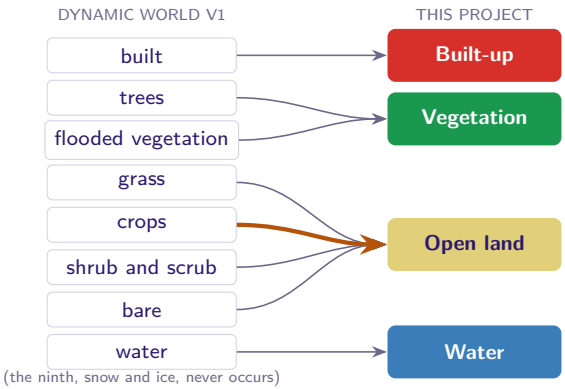


Winter 2024-25 median composite, 16 scenes

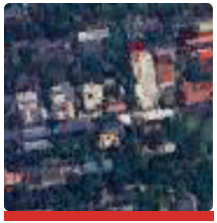


Every dot is one scene that passed the campus cloud test. By April lawns dry out but tree canopy stays green, so the pre-monsoon season helps split them.

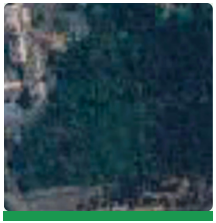
Four campus classes; Dynamic World's eight map onto them



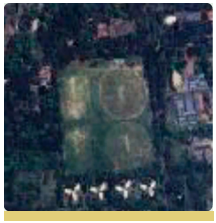
The crops rule. Dynamic World calls about 50 ha of campus lawns *crops*. There is no farmland inside the fence, so crops become open land. Runs 1-3 trained on this remap; the final map uses it only to compare with Dynamic World.



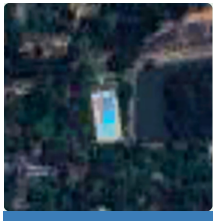
Built-up
roofs, roads, courts



Vegetation
canopy, dense woody



Open land
lawns, fields, bare plots



Water
ponds, tanks

23 features per pixel; the seasonal and texture ones earn their place

10	winter bands B2-B8A, B11, B12
3	winter NDVI, NDBI, NDWI
2	texture: 3×3 std of NDVI, B8
7	pre-monsoon: 4 bands, 3 indices
1	NDVI drop, winter minus pre-monsoon

= 23 features

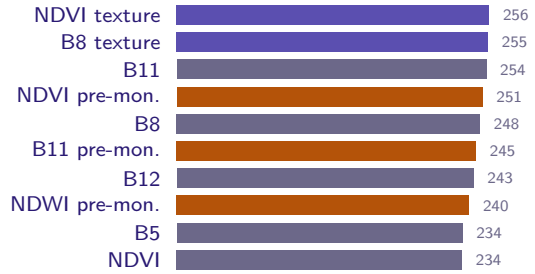
$$\mathrm{NDVI} = \frac{B8 - B4}{B8 + B4}$$

$$\mathrm{NDBI} = \frac{B11 - B8}{B11 + B8}$$

$$\mathrm{NDWI} = \frac{B3 - B8}{B3 + B8}$$

Greenness, built-up and water, each separating a pair of classes that look alike in raw bands.

RANDOM FOREST IMPORTANCE, TOP 10 OF 23 (GINI, EARTH ENGINE)



The **top two are 3×3 texture** of NDVI and NIR, and **5 of the top 10** are seasonal or texture features added in run 3. The ranking is flat: all ten sit within 10% of the top.

Tuning the model: four runs, each fix aimed at one measured error

RUN 1
Baseline
Equal samples per class, one forest per year.

Errors: built-up 224 ha (Dynamic World says ≈110); 91 ha built → vegetation on unchanged roofs; no open-land samples in 2020.

→

RUN 2
Fix the sampling
Samples near class shares (300 / 700 / 150 / 40); **one forest for both years**; crops → open land; 30 m change filter.

Left over: open land 32 ha; points say about 146.

→

RUN 3
Fix open land
Pre-monsoon composite, 3×3 texture, shrub → open land, 450 open samples.

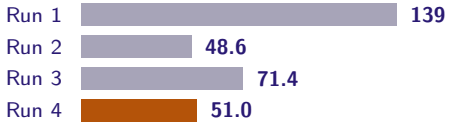
Result: open land 103 ha; OA 0.61 → 0.67 (2020), 0.74 → 0.77 (2025).

→

RUN 4 (FINAL)
Fix the labels
556 points we labelled on WorldView replace Dynamic World as the label source; 500 trees.

Result: open land 156 ha; OA 0.67 → 0.74 (2020), 0.77 → 0.80 (2025).

CHANGED AREA REPORTED (HA)



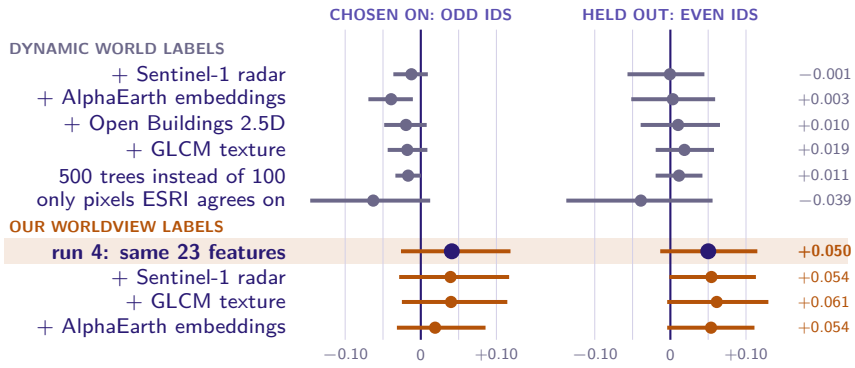
Held-out check, so tuning does not overfit the points. Each change was chosen on the odd-numbered validation points and scored on the even-numbered half it never saw. Mean OA of both years, first version → run 3 → run 4:

odd half (chosen on) 0.65 → 0.685 → **0.725** **even half (held out)** 0.70 → 0.76 → **0.81**

Run 3 raises change to 71.4 ha because more open land is mapped. Run 4 lowers it to 51.0 ha, mostly by dropping built-up ↔ vegetation flips on roofs and canopy (17.7 → 5.0 ha).

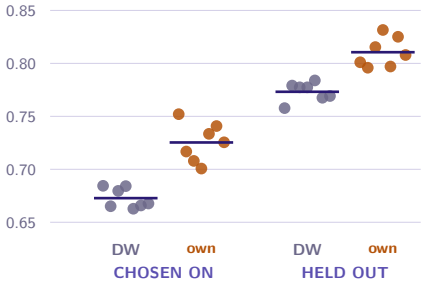
Better labels beat more data

OA GAIN OVER RUN 3, MEAN OF BOTH YEARS, PAIRED 95% CI



Paired against run 3 rerun the same day (OA 0.69 / 0.76), resampled within strata. Each interval includes zero because 365 points are few; the seeds rule out forest luck, not point luck.

SEVEN FORESTS PER LABEL SOURCE, OA



Six random seeds and one 500-tree forest each; bar = mean. **Every forest on our labels beats every forest on Dynamic World's**, on both halves.

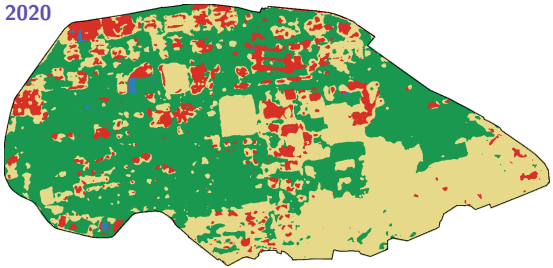
Open land is found again
Producer's accuracy 0.41 → 0.65 (2020) and 0.33 → 0.57 (2025). Open land mapped as built-up: 13 → 7 and 21 → 8 points.

Less like Dynamic World, more right
Agreement with Dynamic World fell (0.82 → 0.79, 0.88 → 0.81) as accuracy on the WorldView points rose. Part of the old agreement was shared error.

What it took
556 points: 260 campus squares and 30 ponds, labelled on the same WorldView scenes, 30 m from every test point. 22 of 24 blind relabels of test points matched.

Vegetation grew, open land shrank: 10.9% of the campus changed

2020

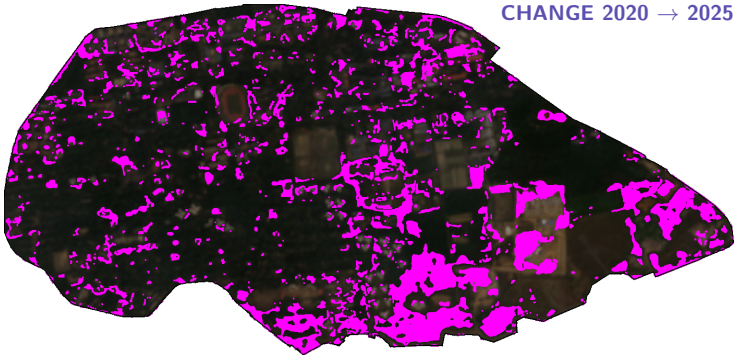


2025

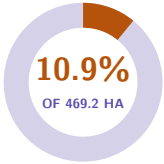


Built-up Vegetation Open land Water

CHANGE 2020 → 2025



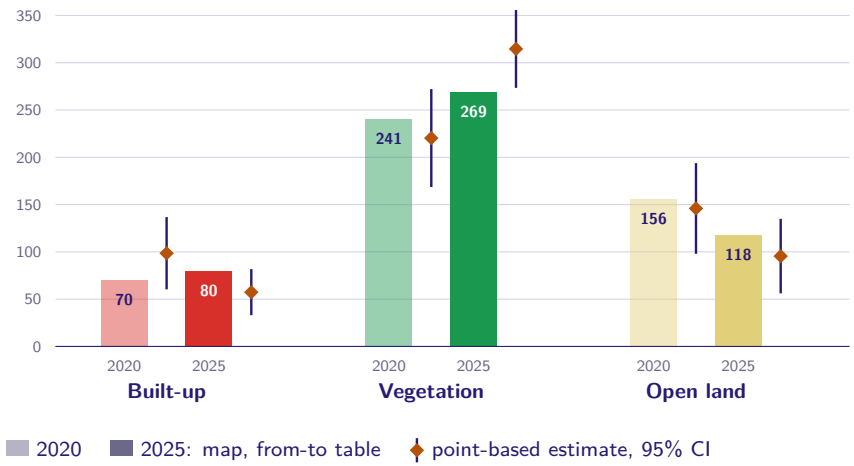
Magenta: changed patches at least 30 m wide, over the 2025 image



51.0 ha CHANGED, ≈10 HA / YEAR

Open land in the south and east turns green, and red at the south-east compound. Same forest, same 23 features in both years, so the difference is the ground, not the model. Before the 30 m width filter the change map holds 91.4 ha; the 40.4 ha it removes is mostly thin edges between classes.

Vegetation +28.7 ha, open land −38.2 ha; the points agree on direction



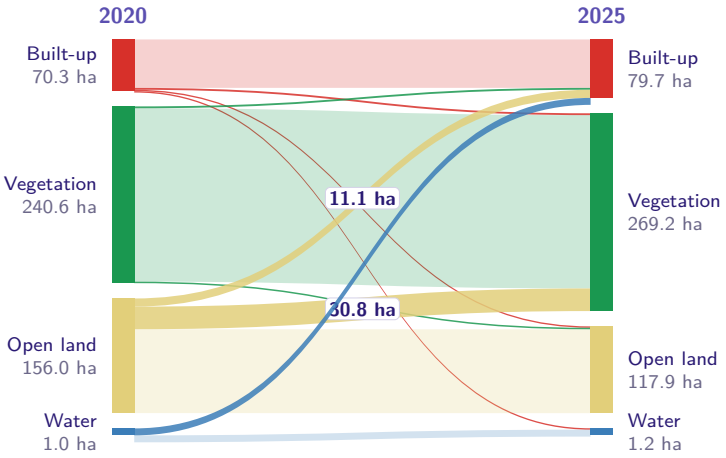
Vegetation +28.7
gained 33.3, lost 4.6 ha

Open land −38.2
gained 3.7, lost 41.9 ha

Built-up +9.4
gained 13.7, lost 4.4 ha

Map vs points. The point estimate (Olofsson et al. 2014) corrects for known map errors. Mapped open land now sits inside its interval in both years (156 vs 146 ha, 118 vs 96 ha); on Dynamic World labels it sat about 40 ha below. Built-up: map +9.4, points −41 ± 45 ha; neither is distinguishable from zero. Only vegetation's intervals do not overlap (169-272 vs 273-356 ha).

Four-fifths of the change is open land turning green or built



Band width = hectares; faint bands stayed the same class. From-to table after the 30 m filter.

ha	gained	lost	net	swap
● Built-up	13.7	4.4	**+9.4**	8.7
● Vegetation	33.3	4.6	**+28.7**	9.2
● Open land	3.7	41.9	**−38.2**	7.4

Swap = 2 × min(gain, loss): area that changed place without changing the class total (Pontius et al. 2004).

30.8 **Open → vegetation**
south and east-end grassland growing into scrub

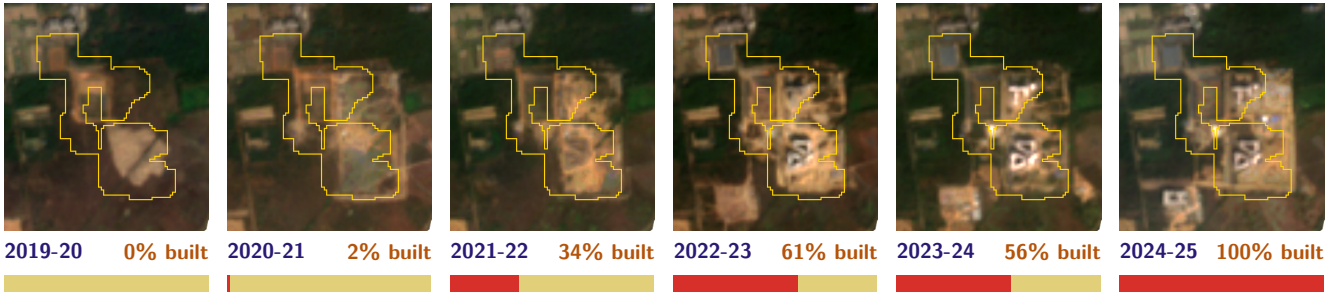
11.1 **Open → built-up**
south-east construction compound

2.5 **Vegetation → built-up**
new blocks in the south

2.5 **Built-up → vegetation**
canopy closing over roofs

Built-up now grows mostly one way. Net +9.4 ha of 18.1 ha changed, 8.7 of it swap. On Dynamic World labels it swapped 29.0 of 33.5 ha, mostly canopy over unchanged roofs: the 10 m pixel, not new ground.

The south-east compound was cleared in 2020 and had roofs by early 2023



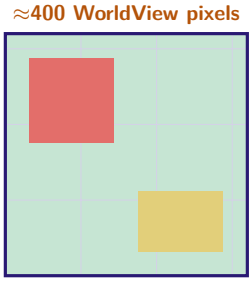
Winter true colour, site in yellow. Bar: the forest's class for the site's open → built-up pixels ● built-up ● vegetation ● open land

Method
The forest trained on 2020 and 2025 classifies every winter in between, with its pre-monsoon pair. Site: the largest open → built-up cluster, 6.2 of the campus's 11.1 ha, inside a 16 ha outline.

Cleared in 2020, roofed by 2023
A bare patch shows by early 2020 and the whole site is bare in 2020-21. Roof shapes appear in 2022-23; by 2024-25 a second block stands to the south-west.

The map runs ahead of the roofs
34% "built" in 2021-22 is bare soil, the roof-like SWIR signal from the accuracy slide. The 2024 dip is year-to-year noise; 0% and 100% at the ends hold by definition.

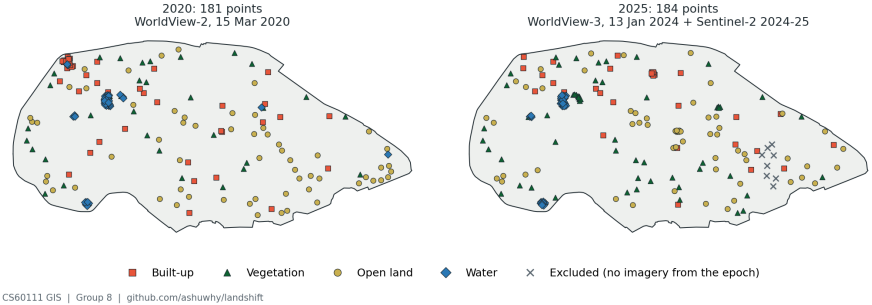
Validation labels come from WorldView, 20 times sharper than Sentinel-2



A 10 m pixel that mixes a roof and a lawn is one label to the classifier but readable by eye on WorldView.

Stratify by map class, 25+ points per class per year → **Label by eye** WorldView-2 (15 Mar 2020), WorldView-3 (13 Jan 2024) → **Offset-correct** to the Sentinel-2 grid → **Area-weight** OA, user's, producer's, area ± 95% CI

Independent validation points, labelled on high-resolution imagery



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365 points: 181 for 2020, 184 for 2025, labelled by Krishnkant Sahu. Never used for training: every training point is at least 30 m away.

Accuracy 0.74 (2020) and 0.80 (2025); built-up is now the weak class

OVERALL ACCURACY 2020

0.74

95% CI 0.64 to 0.84

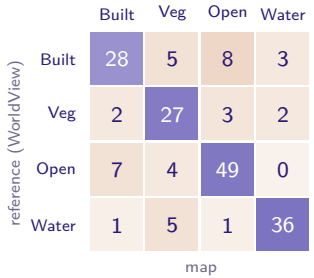
OVERALL ACCURACY 2025

0.80

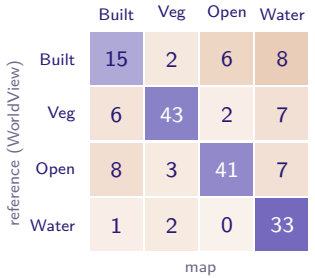
95% CI 0.71 to 0.89

Area-weighted (Olofsson et al. 2014). Against Dynamic World's held-out pixels the map agrees 0.79 / 0.81 (kappa 0.68 / 0.70), down from 0.82 / 0.88 when Dynamic World was the label source. That is agreement with another map, not accuracy.

2020: OA 0.74 (0.64 to 0.84)



2025: OA 0.80 (0.71 to 0.89)



Point counts. Rows: WorldView reference; columns: map. Shade: share of the row; indigo diagonal is correct, saffron is error.

Built-up: producer's accuracy 0.47 in 2020, user's 0.44 in 2025. Roofs under canopy go to vegetation, paved courts to open land. Open land, the weak class on Dynamic World labels, rose from 0.41 / 0.33 to 0.65 / 0.57.

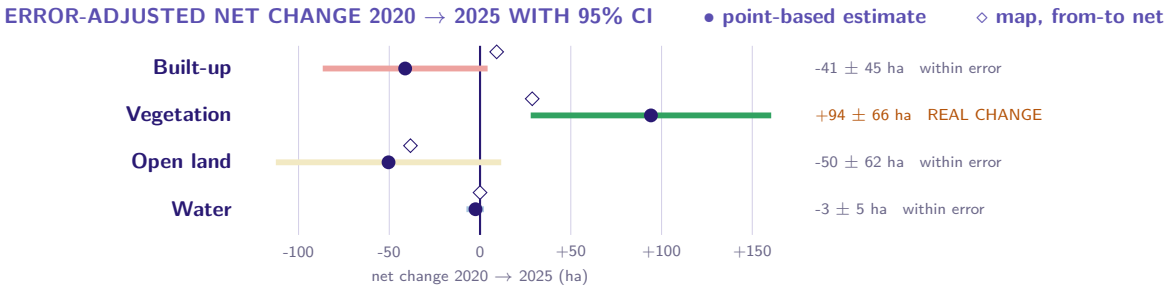
Open → built-up

7 of 60 points in 2020, 8 of 59 in 2025, down from 13 and 21. Bare plots and dry courts share the SWIR signal of roofs.

False water in 2025

22 points mapped water, up from 1 in run 3: shade north of new tall blocks, dark roofs, canopy on pond banks. All sit in the 0.9 ha water stratum, so areas barely move.

Only the vegetation gain is bigger than our map error



Method
Olofsson adjusted area per class, per year, from the WorldView points. The 2020 and 2025 point sets are independent, so the half-width of the difference is $\sqrt{h_{2020}^2 + h_{2025}^2}$.

Vegetation $+94 \pm 66$ ha
The whole interval sits above zero ($z = 2.8$). Campus greening is the one change we can defend against map error.

Built-up and open land
Intervals cross zero (-41 ± 45, -50 ± 62 ha). Halving them needs about $4\times$ the points (CI $\propto 1/\sqrt{n}$), so that is where new labelling should go.

10.9% of the campus changed, mostly open land turning green

51.0 ha

changed: 30.8 open → vegetation, 11.1 open → built-up (south-east compound), 2.5 each way between built-up and vegetation

+28.7 / −38.2 ha

net vegetation and open land after the 30 m filter; built-up +9.4 ha

0.74 / 0.80

overall accuracy on 365 independent WorldView points, up from 0.67 and 0.77

556 points

labelled by eye on WorldView: the change that raised accuracy, where radar, embeddings and texture did not

Yearly composites date the biggest site: cleared in 2020, roofed by 2023.

WHAT THE MAP CANNOT SUPPORT

- **Built-up is the weak class.** Producer's 0.47 in 2020, user's 0.44 in 2025. **So:** read built-up change as approximate.
- **Change accuracy is not measured.** The change map combines two maps at 0.74 and 0.80. **So:** read 51.0 ha as an estimate.
- **10 m pixels mix surfaces.** Roofs under canopy blur; GLCM texture did not help; six bands are 20 m. **So:** small features are unreliable.
- **The 30 m filter drops narrow change.** It cuts 91.4 ha of raw change to 51.0. **So:** a single new building on a lawn can be missed.
- **The label gain is not yet significant.** +0.05 and +0.04 OA; each 95% interval includes zero. **So:** more test points would settle it.
- **Few points, tiny water.** Overall-accuracy intervals are about ±0.10; water is under 2 ha. **So:** water area is not estimable.

WHAT WE WOULD DO NEXT

Shadow fix	Change accuracy	4× the points	3 m imagery
keep shade north of tall blocks out of water	sample the change map itself; change ± CI	halve the built-up and open-land intervals	roofs under canopy, where 10 m texture failed

Who did what

Ashutosh Sharma
23CS10005

- Earth Engine pipeline, first drafts to `gee/lulc_pipeline.js`
- Diagnosed run 1 and ran the tuning: proportional sampling, one forest, crops rule, 30 m filter, pre-monsoon and texture
- Run 4: labelled 556 training points on WorldView; tested them against radar, embeddings and texture
- Area-weighted accuracy and area estimates (Olofsson, Stehman)
- Yearly composites dating the south-east compound
- Python runner that reproduces every number; all figures, README and this deck

Krishnkant Sahu
23CS10035

- Independent validation set: 365 points, stratified on map classes
- Labelled on WorldView-2 and -3, offset-corrected to Sentinel-2
- Labelling notes: dates, rules, hard cases
- Point layers, sampling and export scripts

Sanskar Sovitkar
24CS10131

- Study-area figure: Google Earth Pro view with OSM boundary, north arrow, scale bar
- Literature review of Sentinel-2 / Random Forest change studies
- Framed the research gap: global products vs one campus

Code and results: **github.com/ashuwhy/landshift** · gee/lulc_pipeline.js · scripts/run_pipeline.py · slides/make_charts.py

Thank you

10.9% of the campus changed,
mostly open land turning green.

github.com/ashuwhy/landshift

Ashutosh Sharma
Krishnkant Sahu
Sanskar Sovitkar

